



Lec 5: Word Embeddings

Instructor: Xiang Ren
USC CSCI 544 Applied NLP
Fall 2026

Lecture Outline

- Recap
 - Logistic Regression
 - Model
 - Loss
 - Optimization
 - Regularization
- Multinomial Logistic Regression
- Word Embeddings

Announcements

- Project Group Formation
 - As of now: we have 42 teams of 6 + 4 teams with 4-7 people
 - group 29 (5/6)
 - group 3 (5/6)
 - group 44 (4/6) <-- in contact with each other and would like to proceed
 - group 19 (7/6) <-- in contact with each other and would like to add one person
 - group 41 (7/6) <-- everyone not currently in a group

<https://cloud.google.com/billing/docs/how-to/edu-grants>

Class Project - Proposal

Due on Sep 10 → each team submit one on Brightspace
5% of total grade

Student teams should submit a one-page, single-spaced proposal using 10-point font.

Your project may be either:

Research-focused: Develop methods, models, and analyze data for an existing NLP problem, or formulate a new problem.

Application-focused: Train, fine-tune, or deploy NLP models in a *new application area*. An application-focused project does not necessarily need to introduce a novel research question.

Class Project - Proposal

RUBRIC

Total score: 20

[4 points] Project Idea and Objectives: They must clearly explain what they are trying to do and identify whether the project is research-focused or application-focused.

[3 points] Related Work: They must explain how the problem is addressed today and the limitations of current approaches.

[5 points] Novelty & Impact: They must explain what is new about their approach, why they think it will succeed, who cares, and what difference it could make.

[6 points] Planning & Assessment

- **[2 points] Risk:** They must explain what could go wrong and how they will pivot if necessary.
- **[2 points] Resources:** They must explain what they will need in terms of time and computation and whether this is reasonable for the semester.
- **[2 points] Milestones:** They must identify two milestones toward completing the project.

[2 points] Clarity and Presentation: The proposal should be clear, organized, and easy to understand.

Recap:
Model (Logistic Regression)
+ Loss + Optimization

How to get the right y ?

- For each feature x_i , introduce a weight w_i , which determines the importance of x_i
 - Sometimes we have a bias term, b or w_0 , which is just another weight not associated to any feature
 - Together, all parameters can be termed as $\theta = [w; b]$
- We consider the weighted sum of all features and the bias

$$z = \left(\sum_d w_d x_d + b \right)$$
$$= \mathbf{w} \cdot \mathbf{x} + b$$

If high, $\hat{y} = 1$

If low, $\hat{y} = 0$

How to determine the threshold?

We need probabilistic models!

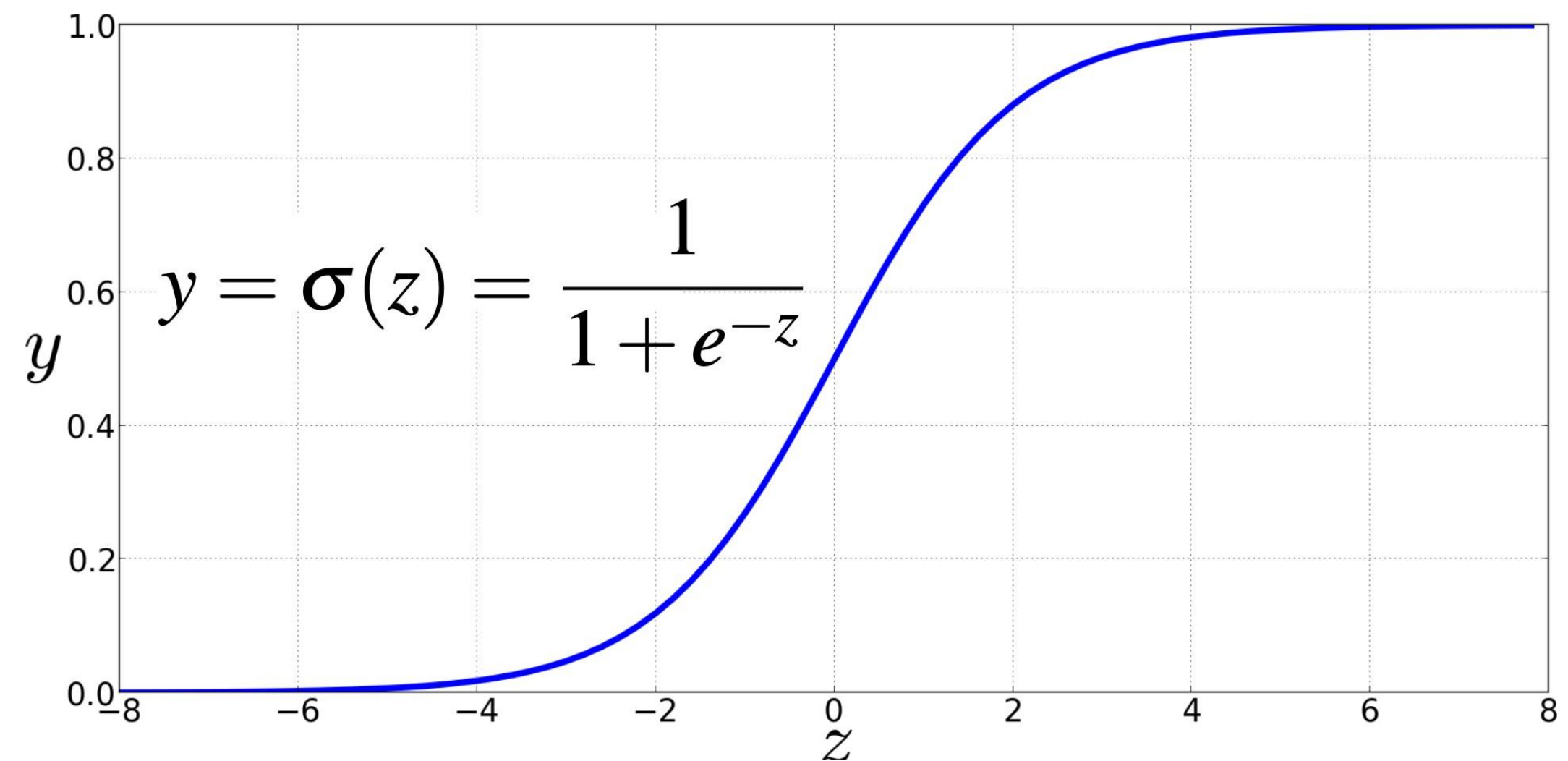
$$P(y = 1 \mid \mathbf{x}; \theta)$$

$$P(y = 0 \mid \mathbf{x}; \theta)$$

Solution: Squish it into the 0-1 range

$$z = \mathbf{w} \cdot \mathbf{x} + b \quad z \in \mathbb{R}$$

- Sigmoid Function, $\sigma(\cdot)$
 - Non-linear!
- Compute z and then pass it through the sigmoid function
- Treat it as a probability!
- Also, a differentiable function, which makes it a good candidate for optimization (more on this later!)



Sigmoids and Probabilities

$$\begin{aligned} P(y = 1|x; \theta) &= \sigma(w \cdot x + b) \\ &= \frac{1}{1 + \exp(-(w \cdot x + b))} \end{aligned}$$

$$\begin{aligned} P(y = 0|x; \theta) &= 1 - \sigma(w \cdot x + b) \\ &= 1 - \frac{1}{1 + \exp(-(w \cdot x + b))} \\ &= \frac{\exp(-(w \cdot x + b))}{1 + \exp(-(w \cdot x + b))} \\ &= \frac{1}{1 + \exp(w \cdot x + b)} \\ &= \sigma(-(w \cdot x + b)) \end{aligned}$$

- Model — Logistic Regression

$$P(y = 1|x; \theta) = \sigma(\boxed{w} \cdot x + \boxed{b}) \quad \text{parameters}$$

- Loss Function: We want to know how far is the classifier output from the true (ground truth / gold standard) label
 - For classification - Cross-entropy loss

$$L(y, \hat{y}) = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))$$

$$L(y, \hat{y}) = - \sum_{i=1}^n y_i \log(\hat{y}_i)$$

The distance between $\hat{y}$ and y

$$\hat{y} = \sigma(\mathbf{w} \cdot \mathbf{x} + b)$$

- We want to know how far is the classifier output from the true (ground truth / gold standard) label

$$y \in \{0,1\}$$

- This difference is called the loss or cost
 - $L(\hat{y}, y)$ = how much $\hat{y}$ differs from y
 - In other words, how much would you lose if you mispredicted
 - Or how much would it cost you to mispredict

Maximizing conditional likelihood

Goal: maximize probability of the correct label $p(y | x)$

Since there are only 2 discrete outcomes (0 or 1) we can express the probability $p(y | \mathbf{x})$ from our classifier (the thing we want to maximize) as

$$p(y | x) = \hat{y}^y (1 - \hat{y})^{1-y}$$

	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	1	0
$y = 1$	0	1

Minimizing negative log likelihood

Goal: maximize probability of the correct label $p(y | \mathbf{x})$

Maximize:
$$\begin{aligned}\log p(y | x) &= \log(\hat{y}^y (1 - \hat{y})^{1-y}) \\ &= y \log \hat{y} + (1 - y) \log(1 - \hat{y})\end{aligned}$$

Now flip the sign for something to minimize (we minimize the loss / cost)

Minimize:
$$\begin{aligned}L_{CE}(y, \hat{y}) &= -\log p(y | x) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})] \\ &= -[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1 - y) \log \sigma(-(\mathbf{w} \cdot \mathbf{x} + b))]\end{aligned}$$

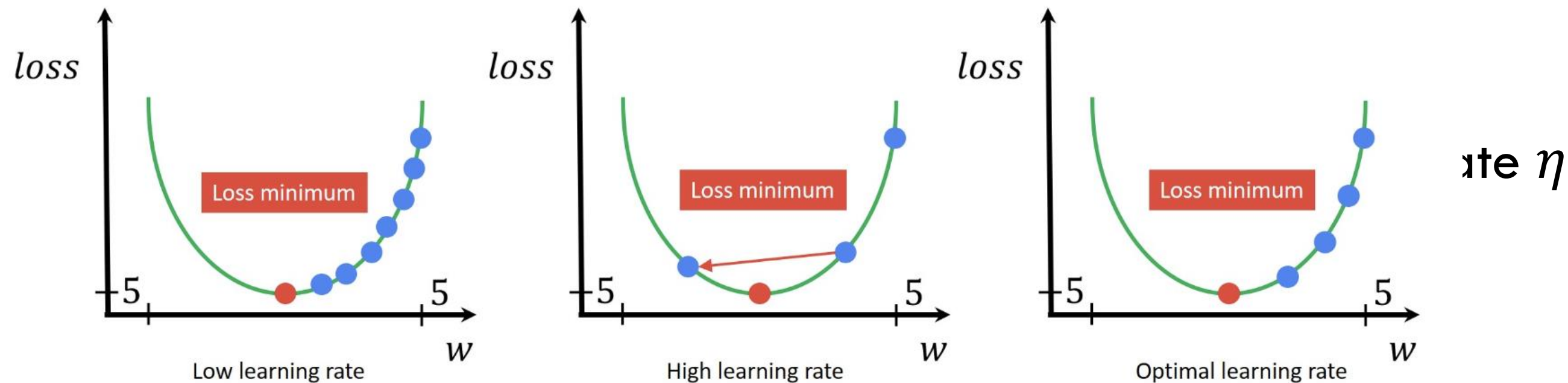
Measures how well the training data matches the proposed model distribution and how good the model distribution is

Cross-Entropy
Loss

Gradient Updates

Move w by the

- Higher learning rate



η Too high: the learner will take big steps and overshoot

$$w_{t+1} = w_t - \eta \frac{\partial}{\partial w} L(f(x; w), y)$$

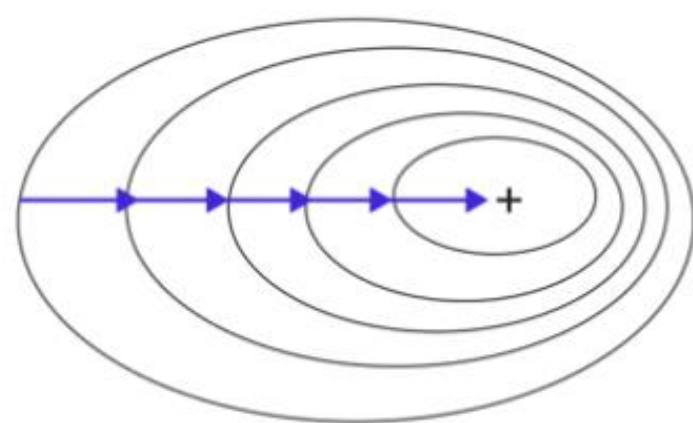
η Too low: the learner will take too long

If parameter w is a vector of d dimensions:

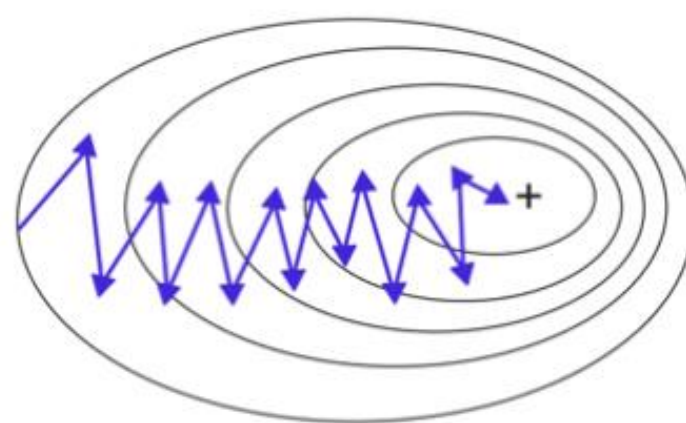
The gradient is just such a vector; it expresses the directional components of the sharpest slope along each of the d dimensions.

- Optimization — How to find the best parameters?
 - Gradient Descent

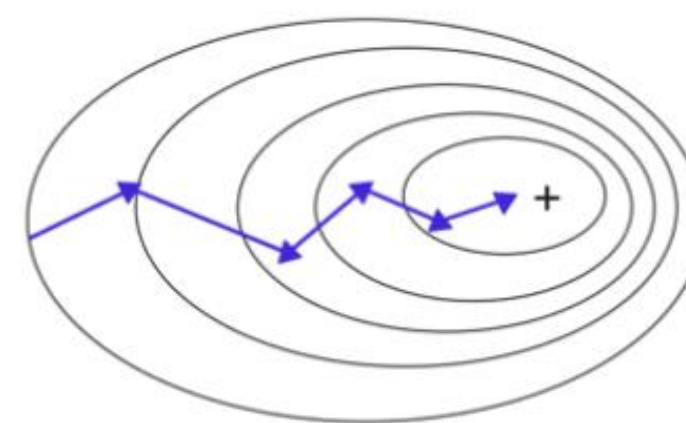
$$w_{t+1} = w_t - \eta \frac{\partial}{\partial w} L(f(x; w), y)$$



GD



SGD



Batch

Pseudocode

function STOCHASTIC GRADIENT DESCENT ($L()$, $f()$, x , y) returns θ

where: L is the loss function

f is a function parameterized by θ

$\mathbf{x}$ is the set of training inputs $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)}$

y is the set of training outputs (labels) $y^{(1)}, y^{(2)}, \dots, y^{(N)}$

$\theta \leftarrow 0$ (or randomly initialized)

repeat till done

for each training tuple $(x^{(i)}, y^{(i)})$: (in random order)

1. Compute $\hat{y}^{(i)} = f(\mathbf{x}^{(i)}; \theta)$

What is our estimated output $\hat{y}^{(i)}$?

2. Compute the loss $L(\hat{y}^{(i)}, y^{(i)})$

How far off is $\hat{y}^{(i)}$ from the true label $y^{(i)}$?

3. $g \leftarrow \nabla L(f(\mathbf{x}^{(i)}; \theta), y^{(i)})$

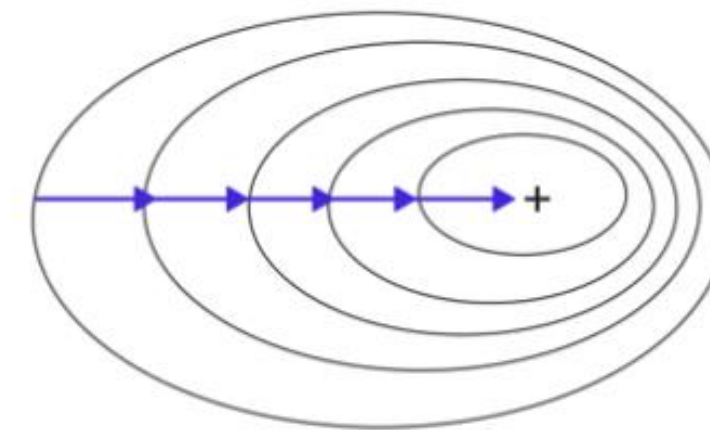
What is the direction that maximizes L ? (gradient)

4. $\theta \leftarrow \theta - \eta g$

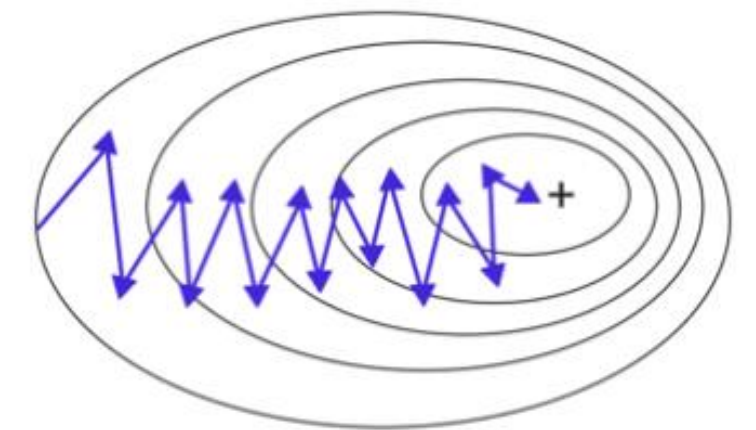
Go the other way instead

return θ

GD



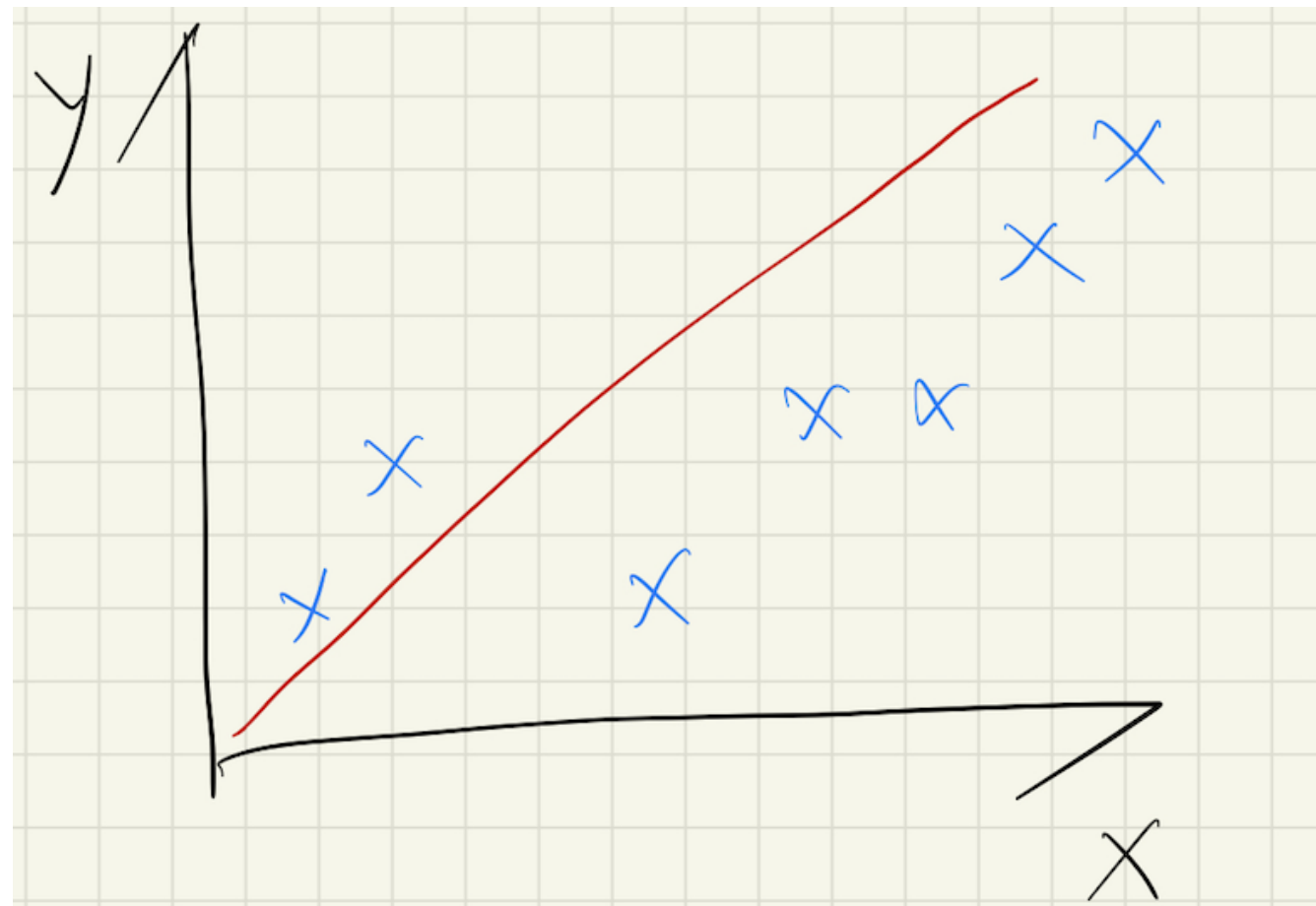
SGD



Stochastic Gradient Descent

Learning: Good or Bad?

avoid this

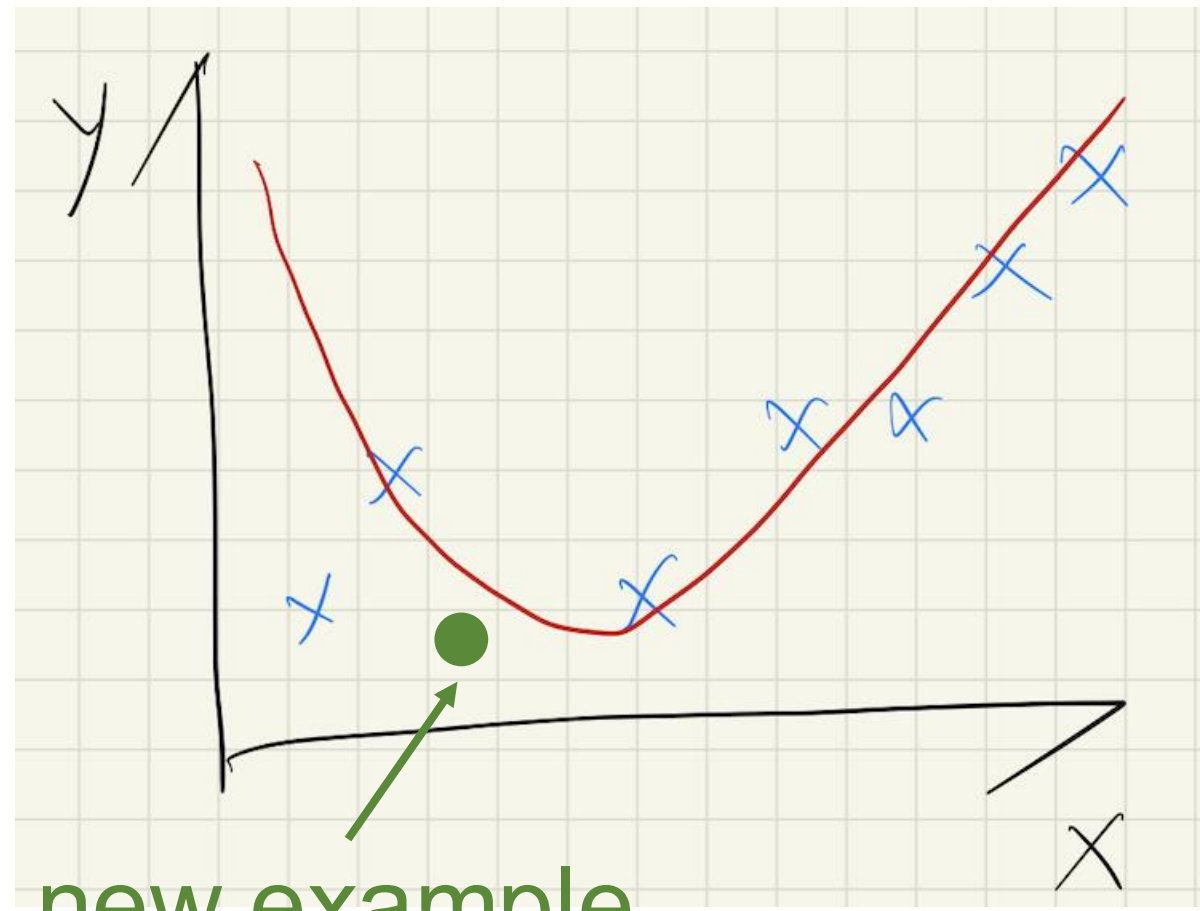


Features = $[1, x]$

underfitting

Too simple

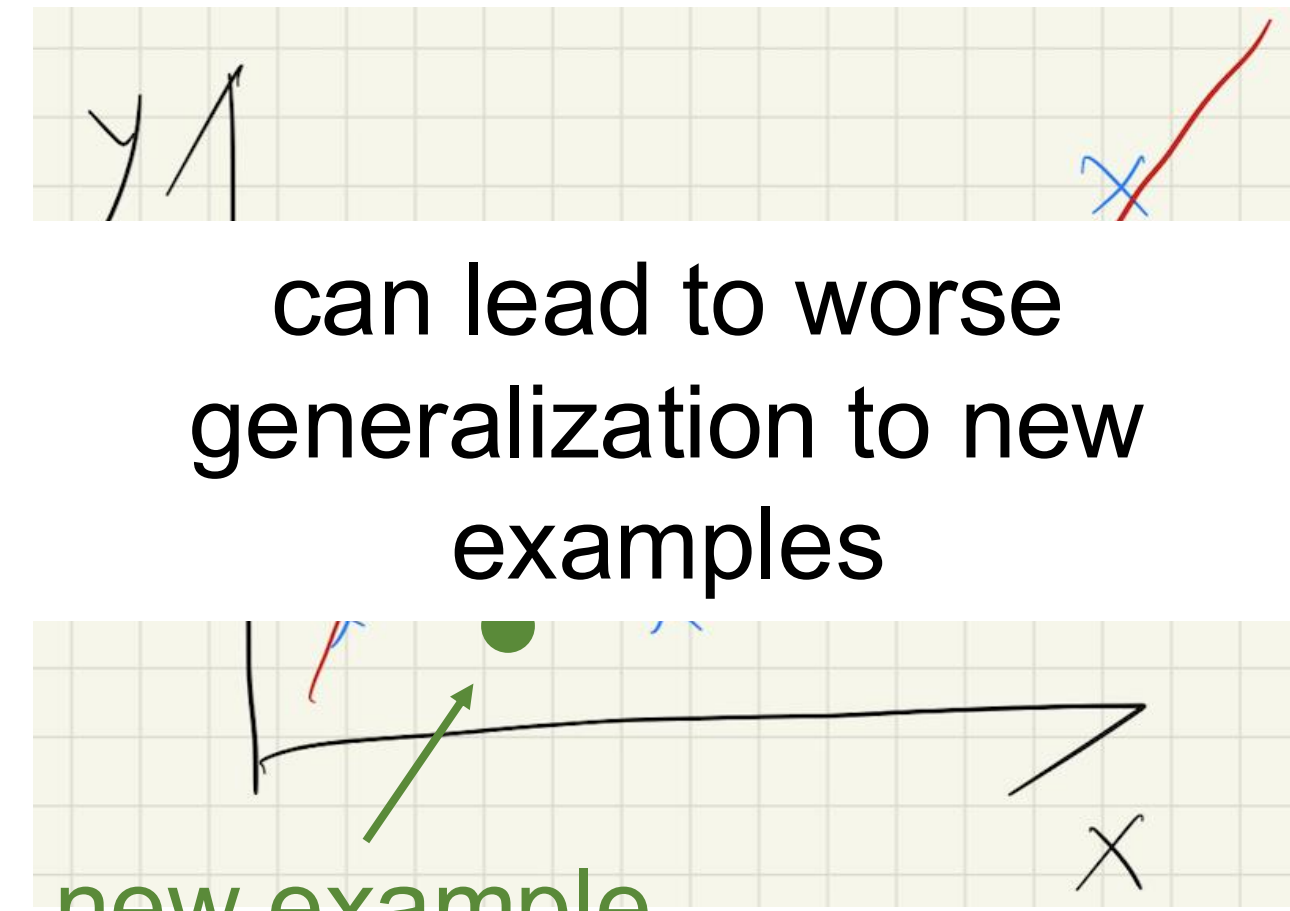
want this



Features = $[1, x, x^2]$

good balance b/w
underfitting and overfitting

avoid this



Features = $[1, x, x^2, \dots, x^7]$

0 training loss!

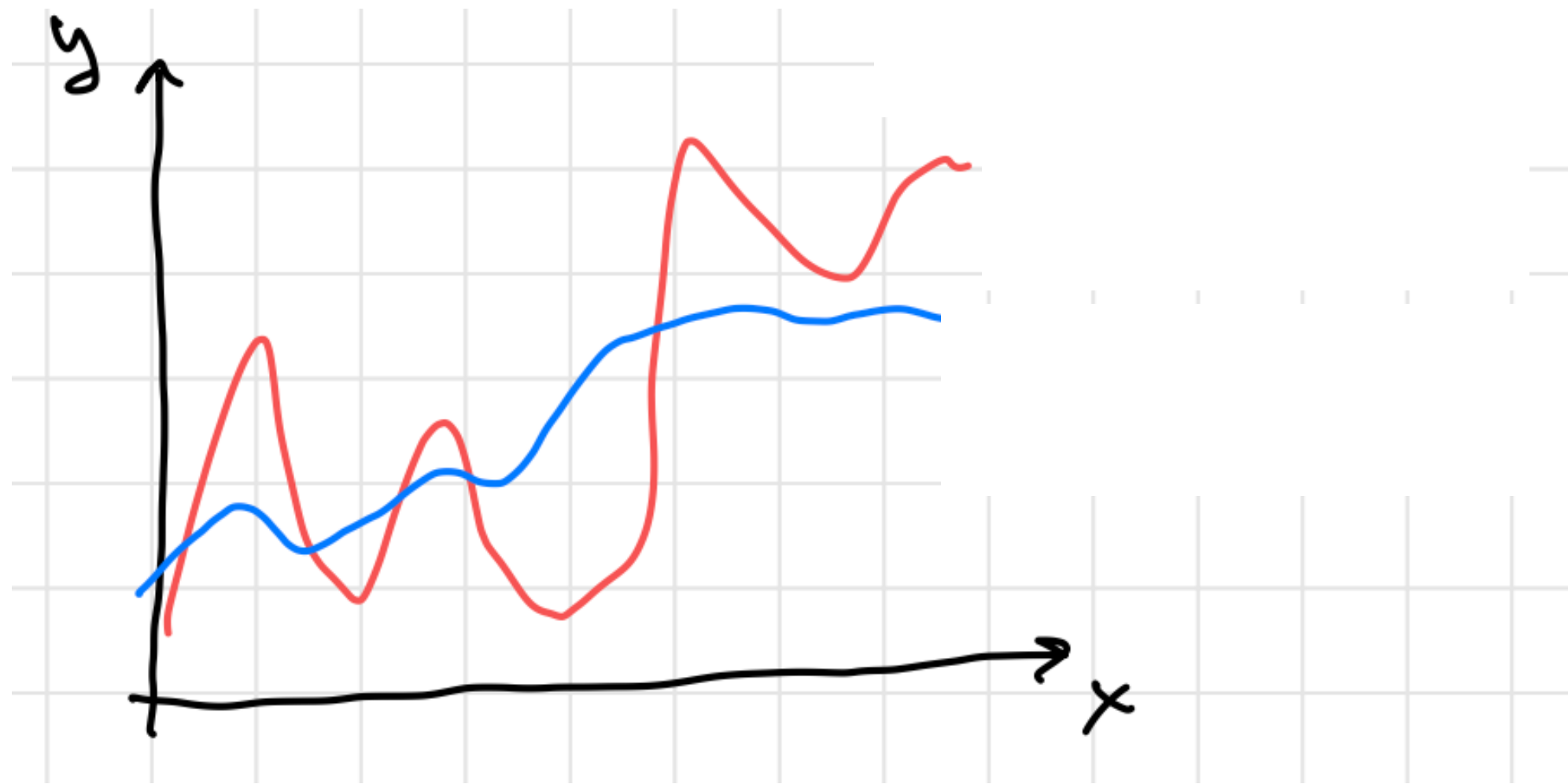
Learns a function that is too
complex

overfitting

Regularization

A technique to reduce overfitting

- by encouraging models to be “simpler”



L2 regularization

Encourage L2 norm of parameters to be small

$$\|w\| = \sqrt{\sum_{j=1}^d w_j^2}$$

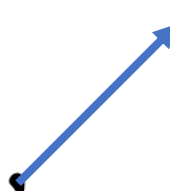
Regularization: How?

$$L(w) = \frac{1}{n} \sum_{i=1}^n (w^\top x^{(i)} - y^{(i)})^2 + \lambda \|w\|^2$$

How does this affect gradient?

$$\nabla_w L(w) = [\text{old gradient}] + \lambda \cdot 2w$$

$\frac{d}{dw_j} \sum_{k=1}^d w_k^2 = 2 \cdot w_j$



- Gradient Descent update: subtract multiple of w from w : i.e., to move towards origin (“weight decay”)

$$w \leftarrow w - \eta([\text{old gradient}] + \lambda 2w)$$

Multinomial Logistic Regression



Multinomial Logistic Regression

- Often we need more than 2 classes
 - Positive / negative / neutral sentiment of a document
 - Parts of speech of a word (noun, verb, adjective, adverb, preposition, etc.)
 - Actionable classes for emergency SMSs
- If > 2 classes we use the term “multinomial logistic regression”
 - Typically the term “logistic regression” indicates binary classification

Multinomial Logistic Regression

The probability of everything must still sum to 1

$$P(+|x) + P(-|x) + P(\sim |x) = 1$$

- Need a generalization of the sigmoid!
- Introducing the softmax function, which
 - Takes a vector $\mathbf{Z} = [z_1, z_2, \dots, z_K]$ of K arbitrary values
 - each z_i corresponds to weighted sum of features for the i -th class
 - Outputs a probability distribution
 - each value in the range $[0,1]$
 - all the values summing to 1

Softmax

The Softmax Function

Turns a vector $\mathbf{z} = [z_1, z_2, \dots, z_K]$ of K arbitrary values into probabilities

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)} \quad 1 \leq i \leq K$$

The denominator $\sum_{j=1}^K \exp(z_j)$ used to normalize all the values into probabilities.

$$\text{softmax}(\mathbf{z}) = \left[\frac{\exp(z_1)}{\sum_{j=1}^K \exp(z_j)}, \frac{\exp(z_2)}{\sum_{j=1}^K \exp(z_j)}, \dots, \frac{\exp(z_K)}{\sum_{j=1}^K \exp(z_j)} \right]$$

Softmax: Example

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)}$$

$$z = [0.6, 1.1, 1.5, 1.2, 3.2, 1.1]$$

$$e^z = [1.8221, 3.0042, 4.4817, 3.3201, 24.5325, 3.0042]$$

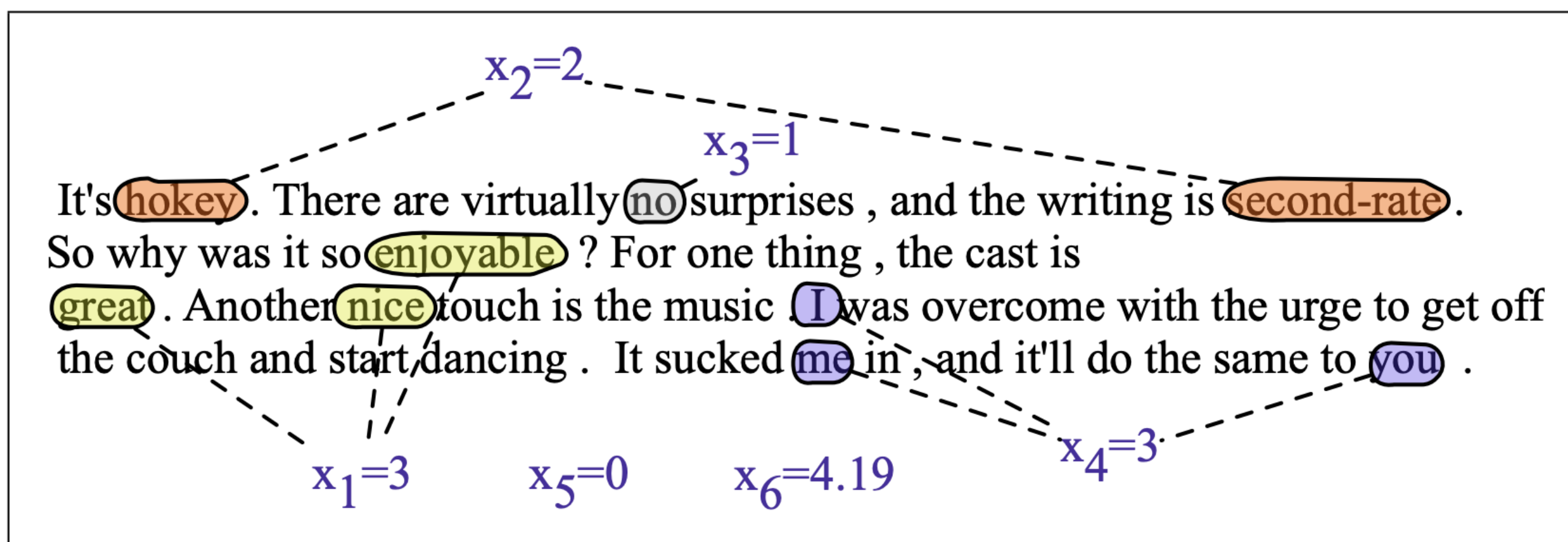
$$\sum_{j=1}^6 e^{z_j} = 1.8221 + 3.0042 + 4.4817 + 3.3201 + 24.5325 + 3.0042 = 40.165$$

$$\text{softmax}(z) \approx [0.0454, 0.0748, 0.1116, 0.0827, 0.6108, 0.0748]$$

Softmax in multinomial logistic regression

$$p(y = c|x) = \frac{\exp(w_c \cdot x + b_c)}{\sum_{j=1}^k \exp(w_j \cdot x + b_j)}$$

- Input is still the dot product between weight vector w_c and input vector x , offset by b_c
- But **separate weight vectors for each of the K classes**
- Parameters are now a matrix $\mathbf{W} \in \mathbb{R}^{K \times d}$ and $\mathbf{b} \in \mathbb{R}^K$

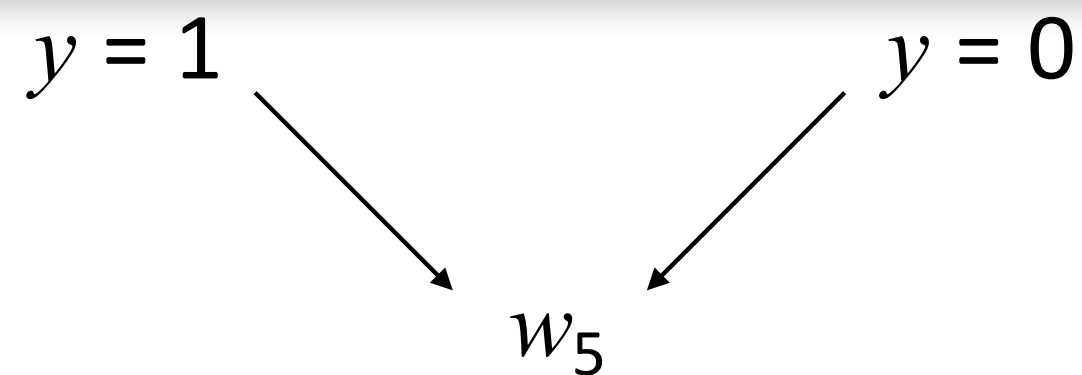


Var	Definition	Value in Fig. 4.2
x_1	count(positive lexicon words $\in$ doc)	3
x_2	count(negative lexicon words $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no" } \in \text{ doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!" } \in \text{ doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	$\ln(\text{word+punctuation count of doc})$	$\ln(66) = 4.19$

Binary versus Multinomial

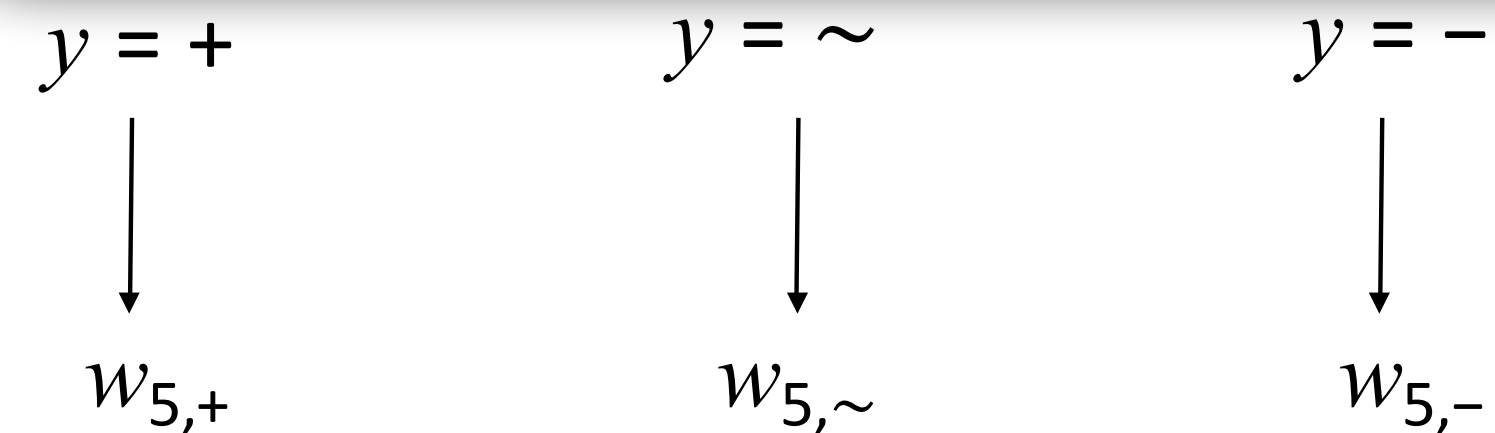
$$x_5 = \begin{cases} 1 & \text{if “!”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$$

Binary Logistic Regression



$$x_5 = \begin{cases} 1 & \text{if “!”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases} \quad w_5 = 3.0$$

Multinomial Logistic Regression



Separate weights for each class

Feature	Definition	$w_{5,+}$	$w_{5,-}$	$w_{5,0}$
$f_5(x)$	$\begin{cases} 1 & \text{if “!”} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	3.5	3.1	-5.3

Loss Function for multinomial logistic regression

label y is a vector with K elements: if $y_c = 1$, the correct label is c ; all other $y_i = 0$

Multinomial LR Loss:

$$L_{\text{CE}}(\hat{\mathbf{y}}, \mathbf{y}) = - \sum_{k=1}^K y_k \log \hat{y}_k \quad p(y_k = 1 | \mathbf{x})$$

Early Neural Language Models

This class: Word Embeddings — the most important component of a neural LM

Case Study: Word Neighborhood Predictor

- Yet do I fear thy nature. It is too full o'the milk of human kindness To catch the nearest way.
- If it were done when 'tis done, then 'twere well It were done quickly.
- Is this a dagger which I see before me, The handle toward my hand?
- My hands are of ;your color, but I shame To wear a heart so white.
- Knock, knock, knock! Who's there,

Where is it more likely to find the word "doth"?



Case Study: Word Neighborhood Predictor

What words are likely to co-occur with the word “garnish”?



Ingredients

12 Large Eggs, Hard-boiled And Peeled
(See Note)

1/2 c. Mayonnaise

2 tbsp. Prepared Yellow Mustard

1/4 tsp. Kosher Salt

[See Nutritional Information](#) ▼

Directions

Slice eggs in half lengthwise. Pop out yolks and place them in a medium-sized mixing bowl. Mash yolks with a fork and then add mayonnaise, mustard, and salt. Mix until smooth. To achieve maximum smoothness, blitz the yolk mixture a few times with an immersion blender. If you are planning to use a pastry bag and piping tip to add the yolk mixture to the egg whites, it helps to get the yolk mixture nice and smooth.

Divide the yolk mixture evenly between the egg white halves. Garnish as desired. Serve immediately or refrigerate until ready to serve.

Word Embeddings

Word: Types & Tokens

“The cat chased the cat.”

Word types: *the, cat, chased* → **3 types**

Word tokens: *the, cat, chased, the, cat* → **5 tokens**

Word: Types & Tokens

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So:

Word = the linguistic item

Type = the category representing identical items

What do words mean?

A **sense** or “concept” is the meaning component of a word

- Can be polysemous (have multiple senses)

Lemmas

- Canonical form
- For example,
break, breaks, broke, broken and breaking all share the lemma “break”
- Different from “stem”

Dictionary

Definitions from [Oxford Languages](#) · [Learn more](#)



ob·jec·tive

/əbˈjektɪv/

Lemma

adjective

1. (of a person or their judgment) not influenced by personal feelings or opinions in considering and representing facts.

"historians try to be objective and impartial"

Similar:

impartial

unbiased

unprejudiced

nonpartisan

disinterested



2. **GRAMMAR**

relating to or denoting a case of nouns and pronouns used as the object of a transitive verb or a preposition.

noun

1. a thing aimed at or sought; a goal.

"the system has achieved its objective"

Similar:

aim

intention

purpose

target

goal

intent

object

end



2. **GRAMMAR**

the objective case.

Sense

Relation: Synonymy

Synonyms: words that have the same meaning in some or all contexts

- couch / sofa
- big / large
- automobile / car
- water / H₂O
- vomit / throw up

Perfect Synonymy might not be possible...

- Even if many aspects of meaning are identical
- Still may differ based on politeness, slang, register, genre, etc.
- e.g. cannot use H₂O in a surfing guide!
- e.g. my big sister != my large sister

Relation: Similarity

Words with similar meanings. Not synonyms, but sharing some element of meaning

- car, bicycle
- cow, horse

word1	word2
vanish	disappear
behave	obey
belief	impression
muscle	bone
modest	flexible
hole	agreement

Human assessment of
word similarity

SimLex-999 dataset
(Hill et al., 2015)

Relation: Antonymy

Senses that are opposites with respect to only one feature of meaning

Otherwise, they are very similar!

- e.g. dark/light, short/long, fast/slow, rise/fall, hot/cold, up/down, in/out

More formally: antonyms can

- define a binary opposition or be at opposite ends of a scale
 - e.g. long/short, fast/slow
- be reversives: denote opposing processes
 - rise/fall, up/down

Relation: Others

Relatedness

- coffee, tea: similar
- coffee, cup: Related, not similar

Connotation

- Words have affective meanings
 - Positive connotations (happy)
 - Negative connotations (sad)

WordNet

WordNet® is a large lexical database of English

Nouns, verbs, adjectives and adverbs are grouped into sets of synonyms (synsets), each expressing a distinct concept

Relations between synsets:

- Super-subordinate relations (hyperonymy, hyponymy or ISA relation)

 - an armchair is a kind of chair, chair is a kind of furniture

- Meronymy (part-of)

 - chair has legs

- Antonymy

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WordNet

A Lexical Database for English

n-grams and Semantics

n-grams do not represent meaning well

Do not tell us that the word “rancor” is close in meaning to the word “hatred”

Or that “Rise” and “Fall” have opposite meanings

Let alone more complex is-a or part-of relations

n-grams and Semantics

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Do not tell us that the word “rancor” is close in meaning to the word “hatred”

Or that “Rise” and “Fall” have opposite meanings

Let alone more complex is-a or part-of relations

Discrete representations of meaning!

Next: feature representations which are continuous

Computational Models of Word Meanings

Can we build a theory of how to represent word meaning, that accounts for at least some of the desiderata?

We will introduce vector semantics

- The standard model in language processing
- Handles many of our goals

“You shall know a word by the company it keeps.”

- Firth (1957)

What does recent English borrowing ongchoi mean?

Suppose you see these sentences:

- Ongchoi is delicious sautéed with garlic.
- Ongchoi is superb over rice
- Ongchoi leaves with salty sauces

And you've also seen theses:

- ... spinach sautéed with garlic over rice
- Chard stems and leaves are delicious
- Collard greens and other salty leafy greens

Conclusion: Ongchoi is a leafy green like spinach, chard, or collard greens

- We could conclude this based on words like 'leaves', 'delicious', and 'sautéed'

Ongchoi: “Water Spinach”

空心菜

Kangkung

Rau muống

...



Yamaguchi, Wikimedia Commons, public domain

Idea 1: Define meaning by linguistic distribution

Let's define the meaning of a word by its distribution in language use, meaning its neighboring words or grammatical environments.

Idea 2: Meaning as a point in space (Osgood et al. 1957)

3 affective dimensions for a word

- Valence: pleasantness
- Arousal: intensity of emotion
- Dominance: the degree of control exerted

	Word	Score		Word	Score
Valence	love	1.000		toxic	0.008
	happy	1.000		nightmare	0.005
Arousal	elated	0.960		mellow	0.069
	frenzy	0.965		napping	0.046
Dominance	powerful	0.991		weak	0.045
	leadership	0.983		empty	0.081

NRC VAD Lexicon
(Mohammad 2018)

Hence the connotation of a word is a vector in 3-space

Idea 1: Define meaning by linguistic distribution

Idea 2: Meaning as a point in multidimensional space

Defining meaning as a point in space based on distribution

Words as Vectors

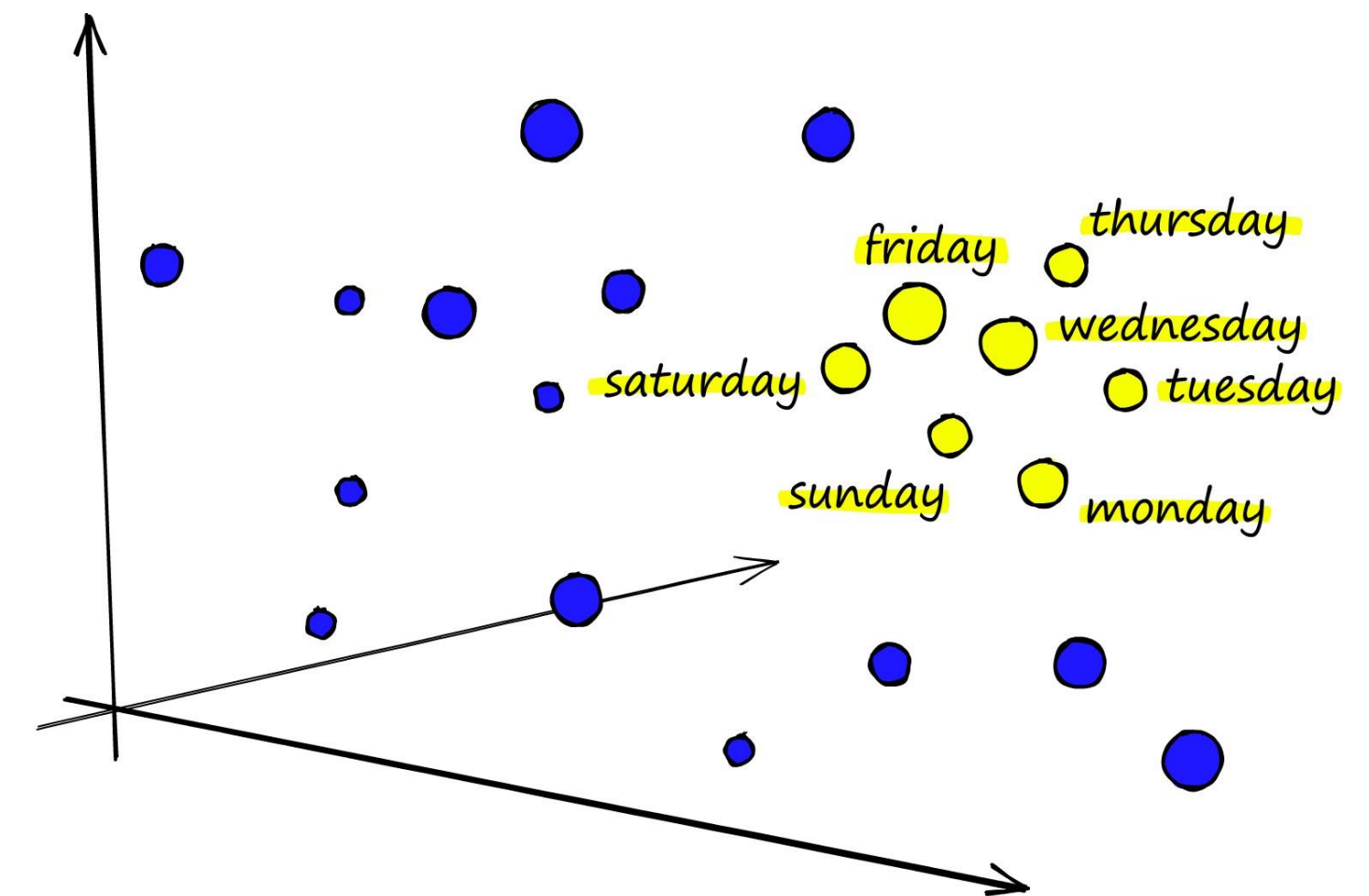
- Each word = a vector
- Similar words are “nearby in semantic space”
- We build this space automatically by seeing which words are nearby in text



Word Embeddings

- Represent a word as a point in a multidimensional semantic space
 - Space itself constructed from distribution of word neighbors
- Called an "embedding" because it's embedded into a space
- Fine-grained model of meaning for similarity

Vector Semantics



Every modern NLP algorithm uses embeddings as the representation of word meaning

Intuition: Why vectors?

Consider sentiment analysis:

- With words, a feature is a word identity
 - Feature 5: The previous word was “terrible”
 - requires exact same word to be in the training and test
- With embeddings:
 - Feature is a vector
 - The previous word was vector [35,22,17...]
 - Now in the test set we might see a similar vector [34,21,14]
 - We can generalize to similar but unseen words!!